DSC 212 - Probability Inequalities

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These inequalities bound tail probabilities like $P(X \geq a)$ using only partial information about X (its mean, its variance, or its MGF). The more information we use, the tighter the bound.

1. Markov's Inequality

Statement: If $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Ex. $E(x) = 10$ $P(X \geq 20) \leq \frac{1}{2}$ "If my average is 10, I can't have too much of my distribution to the right of 20."
If the average is 20 and X is nonnegative then Markov's inequality says the only way for $P(X \geq 20) \geq \frac{1}{2}$ is for $P(X = 20) = \frac{1}{2}$ and $P(X = 0) = \frac{1}{2}$, this give $E(X) = 10$ as the limiting case.

What it needs: only the mean, and X must be *nonnegative*.

Proof: If $X \geq 0$ then $P(X \geq a) \leq \frac{E(X)}{a}$

$$E(X) = \sum_x xP(X = x) \geq \sum_{x \geq a} xP(X = x) \geq \sum_{x \geq a} aP(X = x) = aP(X \geq a)$$

2. Chebyshev's Inequality

Statement: If X has mean μ and finite variance σ^2, then for any $k > 0$,

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

What it needs: the mean and variance. X can be negative.

Proof: Apply Markov to the nonnegative random variable $Y = (X-\mu)^2$ with $b = a^2$:

$$P(|X-\mu| \geq a) = P\big((X-\mu)^2 \geq a^2\big) = \underbrace{P(Y \geq b) \leq \frac{E(Y)}{b}}_{\text{Markov's Inequality}} = \frac{\sigma^2}{a^2}$$

Note: $E(Y) = E[(X-\mu)^2] = Var(X) = \sigma^2$

Application (Weak Law of Large Numbers): Let $X_1, \ldots, X_n$ be iid with mean μ and variance σ^2, and let $\bar{X}_n$ be the sample mean. Then $E[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n\to\infty} 0.$$

3. Chernoff Bound

Statement: For any random variable X with MGF $M_X(t) = E[e^{tX}]$, and any a,

$$P(X \geq a) \leq \frac{M_X(t)}{e^{ta}} = e^{-ta} M_X(t) \qquad \text{for every } t > 0,$$

so the best bound comes from minimizing over t:

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t).$$

For the lower tail, use $t < 0$: $\quad P(X \leq a) \leq \min_{t<0} e^{-ta} M_X(t)$.

What it needs: the whole MGF, which must exist near 0.

Proof: For $t > 0$, the function $x \mapsto e^{tx}$ is positive and increasing, so by Markov

$$P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}.$$

Why it is useful for sums: If $S_n = X_1 + \cdots + X_n$ with the X_i independent, then $M_{S_n}(t) = \prod_i M_{X_i}(t)$, so

$$P(S_n \geq a) \leq \min_{t>0} e^{-ta} \prod_{i=1}^{n} M_{X_i}(t).$$

This gives tail bounds that decay *exponentially* in n, where Chebyshev only gives $1/n$.

Recipe:

1. Write down $M_X(t)$.
2. Form $h(t) = e^{-ta} M_X(t)$, or $\log h(t) = -ta + \log M_X(t)$, which is easier to differentiate.
3. Set $\frac{d}{dt} \log h(t) = 0$, solve for t^*, and check that $t^* > 0$.
4. Plug t^* back in to get the bound.

Example (Normal): $X \sim N(0,1)$, so $M_X(t) = e^{t^2/2}$. For $a > 0$,

$$\log h(t) = -ta + \frac{t^2}{2}, \qquad \frac{d}{dt}\log h(t) = -a + t = 0 \implies t^* = a,$$

$$P(X \geq a) \leq e^{-a^2/2}.$$

For $a = 3$: Chernoff gives $e^{-4.5} \approx 0.011$, while Chebyshev (one-sided, using $P(X \geq 3) \leq P(|X| \geq 3)$) only gives $1/9 \approx 0.111$. The true value is about 0.0013.

Example (Binomial): $X \sim \text{Bin}(n,p)$ with $M_X(t) = (pe^t + q)^n$. Then for $a > np$,

$$P(X \geq a) \leq \min_{t>0} e^{-ta}(pe^t + q)^n.$$

Setting the derivative to zero gives $e^{t^*} = \frac{aq}{(n-a)p}$.

Comparison

Inequality	Uses	Assumption	Tail decay in a
Markov	$E[X]$	$X \geq 0$	$1/a$
Chebyshev	μ, σ^2	finite variance	$1/a^2$
Chernoff	$M_X(t)$	MGF exists	exponential, e.g. $e^{-a^2/2}$